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基本信息

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学科领域

材料物理

微电子科学与工程

集成电路设计与集成系统

材料科学与工程

研究领域

薄膜材料与表面物理

微电子器件与制造

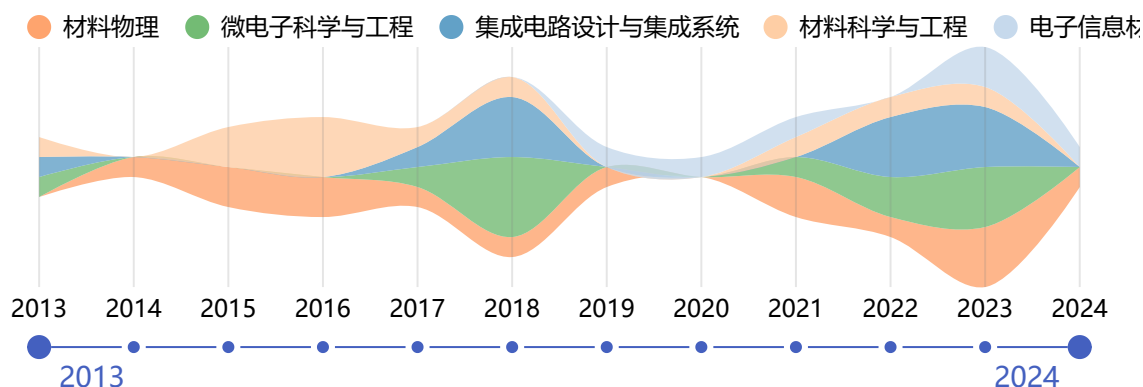
存储器设计与架构

半导体与光电子材料

个人简介

Professor CHOI, BYOUNG DEOG is a distinguished researcher in the field of advanced CMOS technologies, flash memory technology, and display devices. His research interests include device reliability, characterization of MOS capacitors and MOSFETs, gate oxide integrity, and low-temperature polysilicon TFTs. He is affiliated with the CHOI, BYOUNG DEOG Lab at Research & Business Center 7F 85764. His recent publications cover high-performance thin-film transistors, 3-D NAND cell array design, and reliability enhancement of flexible TFTs. He can be reached at +82-31-299-4589 or bdchoi@skku.edu.

近年发表的论文



Facile diffusion of sulfur and fluorine into a-IGTO thin films for high-performance and reliability of transparent amorphous oxide semiconductor thin-film transistors

MATERIALS SCIENCE IN SEMICONDUCTOR PROCESSING

Hong, Jin-Hwan; Kim, Dongbhin; Park, Yubin; Ryu, Jinha; Lee, Saemi; Yoo, Jongmin; Choi, Byoungdeog

被引用次数: 1

2024

String-level compact modeling for dynamic operation and transient analysis of 3D charge trapping flash memory

SOLID-STATE ELECTRONICS

Cho, Sunghwan; Choi, Byoungdeog

被引用次数: 1

2023

Electrical Reliability of Flexible Low-Temperature Polycrystalline Oxide Thin-Film Transistors Under Mechanical Stress

IEEE TRANSACTIONS ON ELECTRON DEVICES

Han, Chanhee; Kim, Hyojung; Kim, Dongbhin; Shin, Jaewoo; Park, Yubin; Byun, Changwoo; Choi, Byoungdeog

被引用次数: 4

2023

Electrical Performance and Reliability Enhancement of a-IGZO TFTs via Post-N₂O Plasma Optimization

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Hyojung; Han, Chanhee; Kim, Dongbhin; Choi, Byoungdeog

被引用次数: 5

2023

Effects of Polycrystalline Silicon Plug Defect on DRAM Characteristics

IEEE TRANSACTIONS ON ELECTRON DEVICES

Han, Dongmin; Hwang, Yoosang; Choi, Byoungdeog

被引用次数: 2

2023

Low-Temperature Processed Complementary Inverter With Tin-Based Transparent Oxide Semiconductors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Chankyu; Hong, Jin-Hwan; Shin, Jaewoo; Lee, Saemi; Kim, Minsoo; Park, Sungsoo; Cho, Junehyeong; Nguyen, Thanh Tien; Choi, Byoungdeog

被引用次数: 0

2023

High-Density 3-D NAND Cell Array Design With Hybrid Bonding

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Seungmin; Lim, Joonsung; Kim, Jun Hyoung; Cho, Sunghwan; Lee, Yong Kyu; Choi, Byoungdeog

被引用次数: 1

2023

Performance and reliability enhancement of flexible low-temperature polycrystalline silicon thin-film transistors via activation-annealing temperature optimization

THIN SOLID FILMS

Kim, Youngrok; Kim, Dongbhin; Ryu, Jinha; Shin, Jaewoo; Lee, Saemi; Choi, Byoungdeog

被引用次数: 0

2023

High-pressure H₂O post-annealing for improving reliability of LTPS and a-IGZO thin-film transistors within a coplanar structure

MATERIALS SCIENCE IN SEMICONDUCTOR PROCESSING

Yoo, Jongmin; Hong, Jin-Hwan; Kim, Hyojung; Kim, Dongbhin; Lee, Chan-kyu; Kim, Minsoo; Byun, Changwoo; Choi, Byoungdeog

被引用次数: 6

2023

Maximizing the Productivity of Photolithography Equipment by Machine Learning Based on Time Analytics

APPLIED SCIENCES-BASEL

Jung, Juyoung; Hong, Jin-Hwan; Suk, Jeewoong; Park, Hyunsoon; Choi, Byoungdeog

被引用次数: 0

2022

Highly Reliable Flexible Device with a Charge Compensation Layer

ACS APPLIED MATERIALS & INTERFACES

Kim, Hyojung; Park, Jongwoo; Khim, Taeyoung; Park, Jungmin; Han, Chanhee; Yoo, Jongmin; Kim, Dongbhin; Song, Jangkun; Choi, Byoungdeog

被引用次数: 4

2022

Charge Pumping Technique to Measure Polarization Switching Charges of FeFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Nam, Kab-Jin; Park, Jung-Min; Choi, Byoung-Deog; Kwon, Kee-Won

被引用次数: 5

2022

A Study on the Role of Pre-Cleaning and a New Method to Strengthen Gate Oxide Quality

NANOMATERIALS

Lee, In-Kyum; Choi, Byoung-Deog

被引用次数: 1

2022

Highly Stackable 3D Capacitor-Less DRAM for a High-Performance Hybrid Memory

IEEE ELECTRON DEVICE LETTERS

Lee, Seungmin; Choi, Byoungdeog

被引用次数: 5

2022

The Performance Enhancement of PMOSFETs and Inverter Chains at Low Temperature and Low Voltage by Removing Plasma-Damaged Layers

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Song, Junhwa; Lee, Eunsun; Hong, Seungho; Kim, Jihun; Oh, Jeonghoon; Choi, Byoungdeog

被引用次数: 0

2022

Reliability Analysis on TiN Gated NMOS Transistors

SCIENCE OF ADVANCED MATERIALS

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Effect of ALD- and PEALD- Grown Al₂O₃ Gate Insulators on Electrical and Stability Properties for a-IGZO Thin-Film Transistor

ELECTRONIC MATERIALS LETTERS

Park, Jungmin; Kim, Hyojung; Choi, Pyungho; Jeon, Bohyeon; Lee, Jongyoon; Oh, Changyong; Kim, Bosung; Choi, Byoungdeog

被引用次数: 18

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Enhanced Hole Injection Characteristics of a Top Emission Organic Light-Emitting Diode with Pure Aluminum Anode

NANOMATERIALS

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被引用次数: 7

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Effect of PECVD Gate SiO₂ Thickness on the Poly-Si/SiO₂ Interface in Low-Temperature Polycrystalline Silicon TFTs

JOURNAL OF ELECTRICAL ENGINEERING & TECHNOLOGY

Park, Jungmin; Choi, Pyungho; Kim, Soonkon; Jeon, Bohyeon; Lee, Jongyoon; Choi, Byoungdeog

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Improved charging phenomenon with a modified barrier structure for flexible displays fabricated on polyimide substrates

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Kim, H. J.; Park, J. M.; Byun, C. W.; Bak, S. R.; Jung, Y. J.; Han, C. H.; Yoo, J. M.; Kim, S. K.; Choi, P. H.; Lee, J. H.; Song, J. K.; Choi, B. D.

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Effects of polyimide curing on image sticking behaviors of flexible displays

SCIENTIFIC REPORTS

Kim, Hyojung; Park, Jongwoo; Bak, Sora; Park, Jungmin; Byun, Changwoo; Oh, Changyong; Kim, Bo Sung; Han, Chanhee; Yoo, Jongmin; Kim, Dongbhin; Song, Jangkun; Choi, Pyungho; Choi, Byoungdeog

被引用次数: 8

2021

Threshold voltage instability and polyimide charging effects of LTPS TFTs for flexible displays

SCIENTIFIC REPORTS

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被引用次数: 20	
The Effects of Valence Band Offset on Threshold Voltage Shift in a-InGaZnO TFTs Under Negative Bias Illumination Stress	
IEEE ELECTRON DEVICE LETTERS	
Kim, Hyojung; Im, Kiju; Park, Jongwoo; Khim, Taeyoung; Hwang, Hyuncheol; Kim, Soonkon; Lee, Sangmin; Song, Minjun; Choi, Pyungho; Song, Jangkun; Choi, Byoungdeog	
被引用次数: 19	2020
A Study on the Charge Trapping Characteristics of High-k Laminated Traps	
IEEE ELECTRON DEVICE LETTERS	
Yoo, Jinhyuk; Kim, Soonkon; Jeon, Woojin; Park, Areum; Choi, Donghee; Choi, Byoungdeog	
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Enhanced Light Extraction from Bottom Emission OLEDs by High Refractive Index Nanoparticle Scattering Layer	
NANOMATERIALS	
Park, Chan Young; Choi, Byoungdeog	
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Analysis of solar cells interconnected by electrically conductive adhesives for high-density photovoltaic modules	
APPLIED SURFACE SCIENCE	
Park, Jisu; Oh, Wonje; Park, Hyungyoul; Jeong, Chaehwan; Choi, Byoungdeog; Lee, Jaehyeong	
被引用次数: 12	2019
Atomic layer-deposited (HfZrO₄)_{1-x}(SiO₂)_x thin films for gate stack applications	
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Choi, Pyungho; Baek, Dohyun; Heo, Sung; Choi, Byoungdeog	
被引用次数: 0	2018
Novel Band-to-Band Tunneling Body Contact (BTBC) Structure to Suppress the Floating-Body Effect in a Vertical-Cell DRAM	
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Cho, Youngseung; Choi, Pyeongho; Hyeon, Younghwan; Song, Junhwa; Hwang, Yoosang; Choi, Byoungdeog	
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New Method for Reduction of the Capacitor Leakage Failure Rate Without Changing the Capacitor Structure or Materials in DRAM Mass Production	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Lee, Jong-Min; Choi, Pyung-Ho; Kim, Soon-Kon; Oh, Jung-Hwan; Shin, Soo-Ho; Noh, Un-Yong; Kim, Hyoung-Sub; Choi, Byoung-Deog	
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Suppression of the Floating-Body Effect of Vertical-Cell DRAM With the Buried Body Engineering Method

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Cho, Youngseung; Kim, Huijung; Jung, Kyungho; Kim, Bongsoo; Hwang, Yoosang; Hong, Hyeongsun; Choi, Byoungdeog

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2018

Effects of Channel Type and Doping on Hysteresis in Low-Temperature Poly-Si Thin-Film Transistors

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Direct evidence of flat band voltage shift for TiN/LaO or ZrO/SiO₂ stack structure via work function depth profiling

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Heo, Sung; Park, Hyoungsun; Ko, Dong-Su; Kim, Yong Su; Kyoung, Yong Koo; Lee, Hyung-Ik; Cho, Eunae; Lee, Hyo Sug; Park, Gyung-Su; Shin, Jai Kwang; Lee, Dongjin; Lee, Jieun; Jung, Kyounggho; Jeong, Moonyoung; Yamada, Satoru; Kang, Hee Jae; Choi, Byoung-Deog

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Novel Approach for the Reduction of Leakage Current Characteristics of 20 nm DRAM Capacitors With ZrO₂-Based High-k Dielectrics

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Lee, Jong-Min; Park, Dong-Sik; Yew, Seung-chul; Shin, Soo-Ho; Noh, Jun-Yong; Kim, Hyoung-Sub; Choi, Byoung-Deog

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2017

High-precision and ultrafast UV laser system for next-generation flexible PCB drilling

JOURNAL OF MANUFACTURING SYSTEMS

Kim, Kwang-Ryul; Cho, Jae-Hee; Lee, Na-Young; Kim, Hyun-Jin; Cho, Sung-Hak; Park, Hong-Jin; Choi, Byoungdeog

被引用次数: 26

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Defect states in amorphous SiN_x:H compounds using thermally stimulated exo-electron emission

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2016

A study of the characteristics of indium tin oxide after chlorine electro-chemical treatment

MATERIALS RESEARCH BULLETIN

Kim, Moonsoo; Kim, Jongmin; Cho, Jaehee; Kim, Hyunwoo; Lee, Nayoung; Choi, Byoungdeog

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2016

Defect visualization of Cu(InGa)(SeS)₂ thin films using DLTS measurement

SCIENTIFIC REPORTS

Heo, Sung; Chung, JaeGwan; Lee, Hyung-Ik; Lee, Junho; Park, Jong-Bong; Cho, Eunae; Kim, KiHong; Kim, Seong Heon; Park, Gyeong Su; Lee, Dongho; Lee, Jaehan; Nam, Junggyu; Yang, JungYup; Lee, Dongwha; Cho, Hoon Young; Kang, Hee Jae; Choi, Pyung-Ho; Choi, Byoung-Deog

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A Study of High-Voltage p-Type MOSFET Degradation Under AC Stress

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Lee, Dongjun; Na, Chungje; Lee, Chiwoo; Lee, Changsub; Hur, Sunghoi; Song, Duheon; Choi, Junghyuk; Choi, Byoungdeog

被引用次数: 3

2015

Ultrafast laser microfabrication of a trapping device for colorectal cancer cells

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Kim, Kwang-Ryul; Kim, Hyun-Jin; Choi, Hyun-Il; Shin, Keong-Sub; Cho, Sung-Hak; Choi, Byoung-Deog

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Direct band gap measurement of Cu(In,Ga)(Se,S)₂ thin films using high-resolution reflection electron energy loss spectroscopy

APPLIED PHYSICS LETTERS

Heo, Sung; Lee, Hyung-Ik; Song, Taewon; Park, Jong-Bong; Ko, Dong-Su; Chung, JaeGwan; Kim, KiHong; Kim, Seong Heon; Yun, Dong-Jin; Ham, YongNam; Park, Gyeong Su; Lee, Dongho; Nam, Junggyu; Kang, Hee Jae; Choi, Pyung-Ho; Choi, Byoung-Deog

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Band alignment of atomic layer deposited (HfZrO₄)_{1-x}(SiO₂)_x gate dielectrics on Si (100)

APPLIED PHYSICS LETTERS

Heo, Sung; Tahir, Dahlang; Chung, Jae Gwan; Lee, Jae Cheol; Kim, KiHong; Lee, Junho; Lee, Hyung-Ik; Park, Gyeong Su; Oh, Suhk Kun; Kang, Hee Jae; Choi, Pyungho; Choi, Byoung-Deog

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Diffusion length and resistivity distribution characteristics of silicon wafer by photoluminescence

MATERIALS RESEARCH BULLETIN

Baek, Dohyun; Lee, Jaehyeong; Choi, Byoungdeog

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2014

Non-uniform integrated Cu(In,Ga)Se₂ solar cell efficiency SPICE simulation

MATERIALS RESEARCH BULLETIN

Back, Doyun; Choi, Byoungdeog; Park, Yongseob; Lee, Jaehyeong

被引用次数: 1

2014

Performance Degradation of c-Si Solar Cells Under UV Exposure

JOURNAL OF NANOSCIENCE AND NANOTECHNOLOGY

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被引用次数: 5

2014

Temperature-Dependent Electrical Characteristics of c-Si and CIGS Solar Cells

JOURNAL OF NANOSCIENCE AND NANOTECHNOLOGY

Choi, Pyung Ho; Baek, Do Hyun; Park, Hyoung Sun; Kim, Sang Sub; Yi, Jun Sin; Kim, Sang Soo; Choi, Byoung Deog

被引用次数: 1

2014

Analysis of the reverse patterning phenomenon caused by a light source change in an attenuated phase shift mask

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Kim, Chulho; Lee, Daeyoup; Kang, Changjin; Chung, Chilhee; Choi, Byoungdeog

被引用次数: 0

2013

Hot hole-induced device degradation by drain junction reverse current

MICROELECTRONICS RELIABILITY

Kim, K. S.; Kim, H. J.; Choi, P. H.; Park, H. S.; Joo, I. H.; Song, J. E.; Song, D. H.; Choi, B. D.

被引用次数: 1

2013

UV laser direct texturing for high efficiency multicrystalline silicon solar cell

APPLIED SURFACE SCIENCE

Kim, Kwang-Ryul; Kim, Tae-Hoon; Park, Hyun-Ae; Kim, Sun-Yong; Cho, Sung-Hak; Yi, Junsin; Choi, Byoung-Deog

被引用次数: 19

2013

学术合作



Hyojung Kim

成均馆大学

合作次数: 9

SONG, JANG KUN

成均馆大学

合作次数: 4

YI JUNSIN

成均馆大学

合作次数: 2

KWON, KEE-WON

成均馆大学

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Nguyen Thanh Tien

成均馆大学

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还有 3 个合作者